VIVEK KUMAR

Backend Software Engineer • Java & Spring Boot • Microservices • AWS Cloud • GenAI

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PROFESSIONAL SUMMARY

Backend Software Engineer with hands-on expertise building resilient, distributed microservices using Java 17, Spring Boot, Apache Kafka, and AWS. Experienced in payment integration, asynchronous message brokers, Redis caching, relational schema optimization, and incorporating GenAI reasoning into backend workflows. Dual AWS Certified: Developer – Associate & Data Engineer – Associate.

TECHNICAL SKILLS

Core & Backend: Java (8/11/17), Spring Boot, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, Microservices, MVC

Cloud & DevOps: AWS (EC2, ECS Fargate, S3, RDS, CloudWatch), Docker, Terraform (IaC), GitHub Actions CI/CD, Linux

Messaging & Caching: Apache Kafka, ActiveMQ, Redis (Caching, Geospatial, Pub/Sub), RabbitMQ

Generative AI & LLMs: LangChain, Spring AI, OpenAI APIs, LCEL, Vector Search, Automated Agent Workflows

Databases & Tools: PostgreSQL, MySQL, MongoDB, Git, Maven, Gradle, JUnit 5, Mockito, Postman, Jira, Agile

PROFESSIONAL EXPERIENCE

Java Developer Intern | HulkHire (Remote) *Jul 2024 – Dec 2024*

- Engineered enterprise microservice modules in Java 17 & Spring Boot with global exception handling, validation frameworks, and RESTful APIs.
- Integrated Trustly Payment Gateway workflows featuring HMAC-SHA256 signature verification, duplicate transaction filters, and idempotency.
- Implemented Redis caching strategies to offload frequent database queries, slashing database load and decreasing latency by 40%.
- Configured ActiveMQ and Kafka topics for reliable asynchronous event delivery across payment and notification services.
- Authored robust unit and integration test suites using JUnit 5 and Mockito, ensuring over 85% code test coverage.
- Participated in production AWS EC2 and RDS PostgreSQL deployments and handled Jira sprint tasks under Agile methodologies.

KEY PROJECTS

Distributed Ride-Hailing Backend Microservice | Java 17, Spring Boot, Apache Kafka, Redis, WebSockets, Docker, AWS *GitHub*

- Architected high-throughput backend services handling ride dispatching, fare calculation, and real-time state synchronization.
- Utilized Redis Geospatial indexes to achieve sub-50ms driver matching and Kafka topic partitions for guaranteed ordered event processing.
- Secured microservice communications with Spring Security, JWT role-based access control, and WebSocket handshake authentication.

AWS ECS Fargate TicketDesk Deployment Infrastructure | AWS ECS Fargate, Terraform, Docker, AWS RDS PostgreSQL, GitHub Actions *GitHub*

- Provisioned secure, production-grade cloud infrastructure using Terraform: VPC, multi-AZ subnets, Application Load Balancer, and ECS Fargate.
- Automated continuous integration and zero-downtime rolling deployments using GitHub Actions and Amazon Elastic Container Registry (ECR).
- Integrated AWS CloudWatch dashboards and health check alarms for proactive uptime and error rate monitoring.

Enterprise GenAI Knowledge Assistant Connector | Java, Spring Boot / Spring AI, Python, LangChain, pgvector, OpenAI API *GitHub*

- Constructed backend connector bridging relational SQL data with semantic vector search via pgvector for context-aware queries.
- Implemented dynamic prompt construction and LLM function calling to synthesize automated reports from database records.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering | DIT University, Dehradun *2021 – 2025*

Certifications: AWS Certified Developer – Associate (DVA-C02) • AWS Certified Data Engineer – Associate (DEA-C01) • Node.js & Express (Coursera) • React Frontend Development (freeCodeCamp) • Agile Project Management (LinkedIn Learning)